

Overview

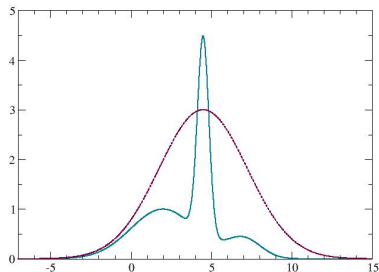
- 1 Monte Carlo integration
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- 2 Introduction: Monte Carlo methods in statistical physics
- 3 Stochastic processes
 - General properties
 - Markov processes
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Quiz this week (in lab)!

Importance sampling

Assume we have a function that is sharply peaked in some region, and close to zero in others

- Need high accuracy where f is big and varying
- Don't waste our time where it is close to zero!



Importance sampling

Sample more points where f is greater, **or**
choose x with probability $p(x)$ so that $f(x)/p(x) \sim \text{constant}$

$$\int f(x)dx = \int p(x) \left[\frac{f(x)}{p(x)} \right] dx$$

Variance is much reduced!

Importance sampling example

Consider the integral

$$I = \int_{-a}^a e^{-x^2/2} dx.$$

This may be evaluated (naively) by picking uniform numbers $X_i \in [-a, a]$ and evaluating

$$I_{MC}^{(naive)} = \frac{2a}{N} \sum_{i=1}^N e^{-X_i^2/2}.$$

The variance of this is given by

$$\text{var } I_{MC}^{(naive)} = \frac{4a^2}{N} \left[\left\langle e^{-X^2} \right\rangle - \left\langle e^{-X^2/2} \right\rangle^2 \right].$$

This gives some significantly nonzero number

Importance sampling example (2)

With importance sampling, the integral becomes

$$I_{MC}^I = Z \int_{-a}^a p(x) dx = Z \langle 1 \rangle_p \quad \text{with} \quad p(x) = \frac{1}{Z} e^{-x^2/2}.$$

where Z is a normalisation constant. The variance is now

$$\text{var } I_{MC}^I = \frac{Z^2}{N} \left[\langle 1^2 \rangle - \langle 1 \rangle^2 \right] = \frac{Z^2}{N} (1 - 1) = 0.$$

In this case, using importance sampling made the integral exact and the variance zero. This is somewhat misleading, since the nontrivial information is now hidden inside the factor Z which we cannot evaluate analytically. A more careful analysis writes the integral as

$$I = \int_{-\infty}^{\infty} \sqrt{2\pi} \Theta(a - |x|) g(x) dx = \sqrt{2\pi} \langle \Theta(a - |x|) \rangle_g,$$

where $g(x)$ is now the ordinary gaussian distribution. Still, in practice the variance is greatly reduced.

Statistical physics

Microscopically

Large number of individual degrees of freedom $\{p_i, q_i\}$, $H(p, q)$

Classical evolution is deterministic and reversible

Macroscopically

A few thermodynamical control variables: T, μ, V

and bulk thermodynamical properties: $\varepsilon, P, S, \dots$

Classical and quantum evolution is statistical and irreversible

Statistical physics: bridge micro—macro

Work with ensembles of microscopic configurations

Probability of configuration $X = \{p_i, q_i\}$ is

$$P(X) \propto e^{-\beta H(X)} \equiv e^{-H(X)/k_B T}$$

Why Monte Carlo?

Only a few systems can be 'solved' analytically

— simulation is an essential tool!

Task

Generate configurations with probability weight $P(X) \propto e^{-\beta H(X)}$

Ideal gas: $H = \sum_i \frac{p_i^2}{2m} \implies P(X) =$ product of gaussians — easy!

With interactions: not so easy!

Common recipe

Construct a series of configurations via a stochastic process

Stochastic processes

This is itself a topic in statistical physics (and mathematics)

'Opposite' of deterministic process (modelled by pdes)

Main topic: **probability** of being in certain states at time t :

- What is $P(x)_t \equiv W_1(x, t)$ — prob of state x at time t ?
- What is $P(x_{t_1}, y_{t_2}) \equiv W_2(x, t_1, y, t_2)$ — prob of x at t_1 **and** y at t_2 ?
- What is $P(x_{t_1}, y_{t_2}, z_{t_3}) \equiv W_3(x, t_1, y, t_2, z, t_3)$ etc?
- What is $P(y_{t_2}|x_{t_1})$ — prob of getting from x at t_1 to y at t_2 ?

Examples

- | | |
|-----------------------|--------------------------------|
| • Brownian motion | • Stock markets |
| • Diffusion and drift | • Many-particle systems |
| • Population dynamics | e.g., Ising model |

Properties of stochastic processes

x, t can be **discrete** or **continuous**

$$\int W_1(x, t) dx = 1;$$

$$W_n \geq 0 \forall n$$

$$\int W_2(x, t_1, y, t_2) dx = W_1(y, t_2); \quad \int W_2(x, t_1, y, t_2) dy = W_1(x, t_2)$$

x can be either a single number (discrete or continuous),
or a **collection** of numbers representing a **configuration**.

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$$\int W_2(x, t_1, y, t_2) dx = W_1(y, t_2); \quad \int W_2(x, t_1, y, t_2) dy = W_1(x, t_2)$$

Stationary processes

$$W_1(x, t) = W_1(x) = P(x) \quad \text{independent of } t$$

$$W_2(x_1, t_1, x_2, t_2) = W_2(x_1, x_2; t_2 - t_1)$$

Ergodic processes

If ensemble mean equals time average then the process is **ergodic**

The process will 'visit' all possible states of the system

- every state must be accessible from every other state
- no periodicity

Autocorrelations

Measure the 'memory' of the process

$$\begin{aligned} R_\phi(t_1, t_2) &\equiv \langle \Delta\phi(t_1)\Delta\phi(t_2) \rangle = \left\langle (\phi(t_1) - \langle\phi(t_1)\rangle)(\phi(t_2) - \langle\phi(t_2)\rangle) \right\rangle \\ &= \int (\phi(x_1, t_1) - \langle\phi(t_1)\rangle)(\phi(x_2, t_2) - \langle\phi(t_2)\rangle) W_2(x_1, t_1, x_2, t_2) dx_1 dx_2 \end{aligned}$$

For stationary process: $\langle\phi(t_1)\rangle = \langle\phi(t_2)\rangle = \text{constant}$

$$R_\phi(t_2 - t_1) = R_\phi(\tau) = \langle \Delta\phi(0)\Delta\phi(\tau) \rangle$$

Typically $R_\phi(\tau) \sim e^{-\tau/\tau_\phi}$, where τ_ϕ is the **autocorrelation time**.

Markov processes

The very simplest type of stochastic process is a **random process**

$$W_2(x_1, t_1, x_2, t_2) = W_1(x_1, t_1)W_2(x_2, t_2) \implies R(\tau) = c\delta(\tau)$$

Next simplest: all information is in W_2 — **Markov process**

Define **transition probabilities**

$$P(x_2 t_2 | x_1 t_1) \equiv T(x_1, t_1 \rightarrow x_2, t_2) \quad \int P(x_2 t_2 | x_1 t_1) dx_2 = 1$$
$$W_2(x_1, t_1, x_2, t_2) = W_1(x_1, t_1)P(x_2 t_2 | x_1 t_1)$$

All information about dynamics of the process is in T .

Where we go next is independent of how we got there.

$$P(x_n t_n | x_{n-1} t_{n-1} \dots x_1 t_1) = P(x_n t_n | x_{n-1} t_{n-1})$$

Only the most recent time counts!

Markov processes

Examples of Markov processes

- random walk
- population dynamics

Examples of non-Markov processes

- self-avoiding random walk
- stock markets

Markov chains

With discrete time steps: **Markov chain**

— need only look at transition probabilities from one time step to next

$$T(Xt_n \rightarrow Yt_{n+1}) \equiv T(X \rightarrow Y)$$

The Master Equation

Look at probability of being in state X at time t , $P(X, t)$

$$P(X, t_{n+1}) = \sum_Y P(Y, t_n) T(Y \rightarrow X)$$

$$P(X, t_n) = \sum_Y P(X, t_n) T(X \rightarrow Y)$$

Master equation

$$P(X, t_{n+1}) - P(X, t_n) = \sum_Y \left[P(Y, t_n) T(Y \rightarrow X) - P(X, t_n) T(X \rightarrow Y) \right]$$

Continuous time:

$$\frac{\partial P(x, t)}{\partial t} = \int \left[P(y, t) w(y, x) - P(x, t) w(x, y) \right] dy$$

Detailed balance

For a **stationary process** $P(X, t_{n+1}) = P(X, t_n) = P(X)$

$\Rightarrow$ rhs of Master Equation is 0.

Sufficient condition

$$P(X)T(X \rightarrow Y) = P(Y)T(Y \rightarrow X) \quad \text{Detailed balance}$$

Can prove that Master Eq describes irreversible process towards equilibrium

Markov chain Monte Carlo

Use such a process to create configurations X
with equilibrium distribution $P(X)$

Summary

Markov processes

- A Markov process is a stochastic process with no memory
- Described by transition probabilities
- All Markov processes obey the master equation
- Detailed balance is a sufficient condition for a stationary process